

Cilindro con CO<sub>2</sub>: presión final, densidad y coeficiente de compresibilidad

Ejercicio 110 %

10 %

✓ Resuelto

El volumen inicial de un cilindro que contiene dióxido de carbono es 0,1 m<sup>3</sup>. La presión absoluta inicial es 100 kPa y la temperatura es 300 K. El gas se comprime de forma isotérmica hasta un volumen de 0,04 m<sup>3</sup>.

- Presión manométrica (dyn/cm<sup>2</sup>) si el gas se comporta como gas ideal.
- Densidad final (UTM/m<sup>3</sup>).
- Coeficiente de compresibilidad cúbico del CO<sub>2</sub> tras ser comprimido, en unidades del SI. Justificar la respuesta.

 Datos

| Variable                      | Valor                                                                   | Notas                            |
|-------------------------------|-------------------------------------------------------------------------|----------------------------------|
| Volumen inicial               | $V_0 = 0,1 \text{ m}^3$                                                 | Cilindro con pistón              |
| Presión absoluta inicial      | $p_{0,\text{abs}} = 100 \text{ kPa}$                                    |                                  |
| Temperatura                   | $T = 300 \text{ K}$                                                     | Proceso isotérmico               |
| Volumen final                 | $V_f = 0,04 \text{ m}^3$                                                | Se comprime hasta este volumen   |
| Presión atmosférica           | $p_{\text{atm}} = 1 \text{ atm} = 101,325 \text{ kPa}$                  |                                  |
| Masa molar CO <sub>2</sub>    | $M = 44 \text{ g/mol}$                                                  |                                  |
| Constante de los gases (dada) | $R = 0,082 \frac{\text{atm} \cdot \text{l}}{\text{mol} \cdot \text{K}}$ | Equiv. a 8,314 J/(mol·K)         |
| Conversiones                  | $1 \text{ Pa} = 10 \text{ dyn/cm}^2$                                    | $1 \text{ UTM} = 9,8 \text{ kg}$ |

 Fórmulas

**Ley de Boyle** (proceso isotérmico,  $T = \text{cte}$ ):

$$p_0 V_0 = p_f V_f \implies p_{f,\text{abs}} = p_0 \cdot \frac{V_0}{V_f}$$

**Ecuación de estado del gas ideal** ( $pV = nRT$ , con  $n = m/M$ , densidad  $\rho = m/V$ ):

$$\rho = \frac{p \cdot M}{R \cdot T}$$

**Coeficiente de compresibilidad cúbico** — definición general:

$$\beta = - \frac{1}{V} \frac{dV}{dp} \Big|_T$$

Para gas ideal isotérmico ( $pV = C = \text{cte}$ ): despejando  $V = C/p \Rightarrow dV/dp = -C/p^2 = -V/p$ , luego:

$$\beta_{\text{gas ideal, isotérmico}} = \frac{1}{p}$$

### Teoría e Hipótesis

**Proceso isotérmico — Ley de Boyle:** Si  $T = \text{cte}$ , para un gas ideal se cumple  $pV = \text{cte}$ . Al reducir el volumen a  $V_f/V_0 = 0,04/0,1 = 0,4$ , la presión absoluta aumenta en proporción inversa:  $p_f = 2,5 p_0$ .

**Unidad Técnica de Masa (UTM):** En el sistema técnico, la UTM es la masa a la que una fuerza de 1 kgf imparte una aceleración de 1 m/s<sup>2</sup>. Como 1 kgf = 9,8 N, se tiene 1 UTM = 9,8 kg.

**Justificación de  $\beta = 1/p$ :** La definición  $\beta = -(1/V)(dV/dp)|_T$  requiere derivar  $V(p)$  a temperatura constante. Para el gas ideal,  $pV = C \Rightarrow V = C/p$ , y la derivada  $dV/dp = -V/p$ , dando  $\beta = 1/p$ . A mayor presión, mayor resistencia a la compresión adicional.

**Comparación:** Para el agua,  $\beta \approx 4,5 \times 10^{-10} \text{ Pa}^{-1}$ , cuatro órdenes de magnitud menor que el CO<sub>2</sub> a 250 kPa. Los gases son mucho más compresibles que los líquidos.

### Resolución

#### A) PRESIÓN MANOMÉTRICA FINAL (DYN/CM<sup>2</sup>)

**¿Por qué?** Se aplica la definición de compresibilidad:  $\Delta p = -K_c \Delta V/V_0$ . La presión manométrica final es la presión relativa respecto a la atmosférica; hay que convertir el resultado a dyn/cm<sup>2</sup> usando los factores de escala del sistema CGS.

**Presión absoluta final — Ley de Boyle ( $T = \text{cte}$ ):**

$$p_{f,\text{abs}} = p_0 \cdot \frac{V_0}{V_f} = 100 \text{ kPa} \times \frac{0,1 \text{ m}^3}{0,04 \text{ m}^3} = 100 \times 2,5 = 250 \text{ kPa}$$

**Presión manométrica (relativa a la atmósfera):**

$$p_{f,\text{man}} = p_{f,\text{abs}} - p_{\text{atm}} = 250 - 101,325 = 148,675 \text{ kPa} = 148\,675 \text{ Pa}$$

**Conversión a dyn/cm<sup>2</sup> (1 Pa = 10 dyn/cm<sup>2</sup>):**

$$p_{f,\text{man}} = 148\,675 \times 10 = 1\,486\,750 \text{ dyn/cm}^2$$

$$\boxed{p_{f,\text{man}} = 1486,75 \times 10^3 \text{ dyn/cm}^2}$$

### B) DENSIDAD FINAL (UTM/M<sup>3</sup>)

**¿Por qué?** La densidad aumenta al comprimir el fluido:  $\rho_f = m/V_f$  donde  $V_f = V_0 + \Delta V$ . La masa se conserva, solo cambia el volumen. La conversión a UTM/m<sup>3</sup> (unidades técnicas) requiere dividir entre  $g = 9,8 \text{ m/s}^2$ .

Aplicamos  $\rho = pM/(RT)$  con  $R = 0,082 \text{ atm}\cdot\text{l}/(\text{mol}\cdot\text{K})$  tal como da el enunciado. Convertimos  $p_f$  a atm:

$$p_{f,\text{abs}} = \frac{250 \text{ kPa}}{101,325 \text{ kPa/atm}} = 2,467 \text{ atm}$$

$$\rho_f = \frac{p_f \cdot M}{R \cdot T} = \frac{2,467 \text{ atm} \times 44 \text{ g/mol}}{0,082 \frac{\text{atm}\cdot\text{l}}{\text{mol}\cdot\text{K}} \times 300 \text{ K}} = \frac{108,55}{24,6} = 4,41 \frac{\text{g}}{\text{l}} = 4,41 \frac{\text{kg}}{\text{m}^3}$$

**Conversión a UTM/m<sup>3</sup>** (1 UTM = 9,8 kg):

$$\rho_f = \frac{4,41 \text{ kg/m}^3}{9,8 \text{ kg/UTM}} = \boxed{0,45 \text{ UTM/m}^3}$$

### C) COEFICIENTE DE COMPRESIBILIDAD CÚBICO – JUSTIFICACIÓN

**¿Por qué?** El coeficiente de compresibilidad cúbica  $\beta = -\frac{1}{V} \frac{dV}{dp}$  es el recíproco del módulo de elasticidad volumétrico  $K_e$ . Para un líquido,  $\beta$  es muy pequeño (del orden de  $10^{-9} \text{ Pa}^{-1}$ ), lo que justifica la hipótesis de incompresibilidad en la mayoría de los problemas de fluidos.

El coeficiente de compresibilidad cúbico se define como:

$$\beta = -\frac{1}{V} \frac{dV}{dp} \Big|_T$$

**Derivación para gas ideal isotérmico:** A temperatura constante, la Ley de Boyle impone  $p \cdot V = C = \text{cte}$ . Despejando  $V$ :

$$V = \frac{C}{p} \implies \frac{dV}{dp} = -\frac{C}{p^2} = -\frac{pV}{p^2} = -\frac{V}{p}$$

Sustituyendo en la definición de  $\beta$ :

$$\beta = -\frac{1}{V} \cdot \left(-\frac{V}{p}\right) = \frac{1}{p}$$

Evalando a la presión final  $p_f = 250 \text{ kPa} = 250\,000 \text{ Pa}$ :

$$\beta = \frac{1}{p_f} = \frac{1}{250\,000 \text{ Pa}} = \boxed{4 \times 10^{-6} \text{ Pa}^{-1}}$$

El coeficiente de compresibilidad del gas ideal disminuye al aumentar la presión: a mayor presión, mayor resistencia a seguir comprimiendo.

### ✓ Conclusiones

#### RESULTADOS

a)  $p_{f,\text{man}} = 1486,75 \times 10^3 \text{ dyn/cm}^2$

b)  $\rho_f = 0,45 \text{ UTM/m}^3 (= 4,41 \text{ kg/m}^3)$

c)  $\beta = 1/p_f = 4 \times 10^{-6} \text{ Pa}^{-1}$

**Clave:** El proceso isotérmico implica  $pV = \text{cte}$  (Boyle). La densidad se calcula con las unidades del enunciado ( $R = 0,082 \text{ atm}\cdot\text{l}/(\text{mol}\cdot\text{K})$ ). El coeficiente  $\beta = 1/p$  se justifica derivando  $V(p)$  de  $pV = C$ .

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to determine the relationships between the variables.

The results of the study show that there is a significant positive relationship between the variables. This finding is consistent with the previous research on the topic. The study also found that there are some differences in the results between the different groups of participants.

The conclusions of the study suggest that the findings have important implications for practice and policy. Further research is needed to explore the relationships between the variables in more detail.

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to identify patterns and trends. The results of the study indicate that there is a significant relationship between the variables being studied.

The findings of the study have several implications for practice and policy. First, the results suggest that the current approach to the issue is not effective. Second, the study highlights the need for further research in this area. Finally, the findings provide valuable insights for the development of new interventions and policies.

In conclusion, the study has provided a comprehensive overview of the research topic. The findings are significant and have important implications for the field. Further research is needed to explore the underlying mechanisms and to develop more effective interventions.

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The results of the study show that there is a significant positive correlation between the variables. This suggests that as one variable increases, the other variable also tends to increase. The findings have important implications for the field of study and may lead to further research in this area.

In conclusion, the study has provided valuable insights into the relationship between the variables. The findings suggest that there is a need for further research to explore this relationship in more detail.

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The research was conducted in a systematic and rigorous manner, following the principles of scientific inquiry. The data was collected from a representative sample of the population, and the analysis was performed using advanced statistical techniques. The results of the study indicate that there is a significant relationship between the variables under investigation, and this finding has important implications for the field of study.

In conclusion, the study has provided valuable insights into the subject matter and has contributed to the existing body of knowledge. The findings suggest that further research is needed to explore the underlying mechanisms and to test the generalizability of the results. The researcher hopes that this work will serve as a foundation for future studies and that it will contribute to the advancement of the field.

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The conclusions drawn from the study are based on the evidence presented in the data. They provide a comprehensive overview of the subject matter and offer valuable insights into the field. The suggestions for future research are based on the limitations of the current study and aim to address the gaps in the existing knowledge.

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The research was conducted in a systematic and rigorous manner, following the principles of good research practice. The data were collected from a representative sample of the population, and the analysis was carried out using appropriate statistical methods. The results of the study are presented in a clear and concise manner, and the conclusions are based on the evidence gathered.

The findings of the study have important implications for the field of research, and they provide valuable insights into the issues being studied. The research also highlights the need for further investigation in this area, and it offers suggestions for how this can be achieved.

In conclusion, the research has shown that there is a need for further investigation in this area, and it has provided valuable insights into the issues being studied. The findings of the study have important implications for the field of research, and they offer suggestions for how this can be achieved.

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